COMP 018: Database Administration - Activity 4 Progress Report

Access Control, Database Users, Roles, Privileges, and Security Governance

Project Identification & Metadata

Course	COMP 018 - Database Administration
Evaluating Group	Group 4: Olivarez, Abangan, Pocsidio, Infantado, and Bartolata
Evaluating Group Repository	https://github.com/Nycri/brewpos
Target Project Monitored	Campus Canteen Pre-Order and Queue Management System
Target Project Group	Group 8 (bwisIT Solutions): Joshua Borreta, Martin Lawrence Bola, Joaquin Phillip Juatchon, Karl Raphael Marcos, Sancho Pacheco Jr, Euan Joseph Vera Cruz
Target Project Repository	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System
Primary Database	canteen_db (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC)
Restored Test Database	canteen_db_test_restore

ACTIVITY 4: Database Users, Roles, Privileges, and Access Control

1.1 Access-Control Plan & Architecture

In default database setups, web backend applications frequently connect using an administrative database owner account. Connecting live applications under an administrative user violates the Principle of Least Privilege because the backend maintains full data definition and unconstrained data manipulation capabilities, creating security vulnerabilities where software flaws could allow unauthorized record truncation or schema alteration.

To address this risk in Group 8's Campus Canteen System, a role-based access control architecture was implemented to decouple administrative database ownership from daily operational traffic. The access structure defines non-login group roles to serve as privilege templates alongside individual login accounts

assigned to their corresponding group roles.

Under this security plan, the application runtime account is granted selection, insertion, and update permissions on operational tables, while its row deletion capabilities are strictly confined to menu item management. The account is prohibited from deleting customer orders, order items, student accounts, queue tokens, or system notifications. Additionally, a dedicated read-only auditor account is granted query access across all tables to support sales analytics and queue inspections without write permissions, while database administrative duties remain isolated under a dedicated owner account.

1.2 Access-Control Configuration Summary

The configuration process established group roles representing runtime and read-only audit profiles, followed by the creation of matching login accounts with secure credentials. Access privileges were revoked from public roles across the database and public schema to enforce explicit access controls. Connection permissions and schema usage rights were granted to the designated group roles, explicit table-level grants were applied according to operational roles, and sequence usage rights were configured to support auto-incrementing key generation.

1.3 Verification Results: Runtime Account Testing

Verification testing confirmed that the application runtime account successfully reads active vendor listings, inserts new menu items while advancing sequence counters, and deletes unreferenced menu items. When deletion operations were attempted on customer order transaction logs, the database engine denied execution under the narrow deletion policy. Data definition commands, such as dropping notification tables, were rejected due to missing table ownership permissions.

1.4 Verification Results: Read-Only Auditor Account Testing

Inspection testing demonstrated that the read-only auditing account successfully executes multi-table reporting queries joining customer orders, student details, vendor stalls, and queue entries. Attempted modification operations, including inserting new vendor records, updating student role privileges, or deleting menu items, were uniformly blocked by the database engine.

1.5 Verification Results: Cross-Database & Administrative Boundary Testing

Administrative boundary testing confirmed that database creation commands attempted under application accounts were denied due to missing cluster-level

administrative privileges. Public connection privileges were revoked, confining runtime accounts strictly to authorized database targets.

1.6 Issues, Fixes, and Feedback to Owners

During local deployment, an environment conflict occurred on network port 8080 due to a pre-existing local service. This was resolved by remapping the backend service port and updating client configuration settings. Least-privilege roles were created to replace default administrative connection settings.

Feedback provided to Group 8 commended the clean, idempotent migration scripts and transactional advisory locks. Recommendations advised configuring dynamic port bindings in application property files and updating default environment examples to reference least-privilege runtime accounts instead of administrative owner credentials.

CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 4)

Column Header	Value
Project	Group 8 - bwisIT Solutions: Campus Canteen Pre-Order and Queue Management System
Repository cloned	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System, Branch: main
Database setup result	Success (Port 8080 collision with local PEMHTTPD identified; resolved by port remapping to 8081). All 7 tables and seed data verified.
Access-control folder link	https://github.com/Nycri/brewpos/tree/main/access-control
Access-control status	Completed
Feedback sent to owners	(04/11/2026)
PDF progress document	Activity4_Progress_Report_Group8.pdf
Last updated	2026-11-04

AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All

database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.